

Emergency Vehicle Sound Classification for Dynamic Traffic Signal Control using Convolutional Neural Networks

Dr.M.Annalakshmi
Computer Science and Engineering
Velammal College of Engineering and
Technology
Madurai, India
annam.baluss@gmail.com

Keerthana K S
Computer Science and Engineering
Velammal College of Engineering and
Technology
Madurai, India
21cse015keerthana@gmail.com

Sowmiya Eswari P
Computer Science and Engineering
Velammal College of Engineering and
Technology
Madurai, India
21cse029sowmiyaeswari@gmail.com

Abstract—Urban traffic management systems must ensure quick and efficient passage for emergency vehicles like fire trucks, ambulances and police cars to reduce response time during critical situations. This paper proposes an approach for accurately classifying emergency vehicle sirens using Convolutional Neural Networks (CNNs). The system utilizes a dataset of WAV audio files representing the sounds of ambulances, fire trucks, and police sirens. Mel-Frequency Cepstral Coefficients (MFCCs) were extracted as key features from the audio files to capture the essential characteristics of the sirens. Preprocessing steps included noise reduction and feature scaling to ensure the model's accuracy. A CNN was then trained to classify the audio into the three categories, achieving over 90% accuracy on the test set. This high level of accuracy demonstrates the system's potential for real-world application in intelligent traffic management systems, where the model can dynamically adjust traffic signals, providing priority access to emergency vehicles in real-time. Future extensions of this work will include integrating the system with live traffic data to improve the overall efficiency of urban mobility.

Keywords— Emergency vehicle classification, CNN, MFCC, sound recognition, AI traffic control, real-time traffic management.

I. INTRODUCTION

Urban traffic congestion has developed into a frequent problem nowadays across cities worldwide, severely impacting the overall efficiency of transportation systems.[1] The growing volume of vehicles on the road leads to increased travel times, fuel consumption, and environmental pollution due to idling traffic. The National Crime Records Bureau estimates that 24,012 individuals die every day because of improperly delayed medical care, often due to the slowing of emergency vehicles[2]. One of the most critical consequences of traffic congestion is the delay it causes for emergency vehicles, including ambulances, fire trucks, and police cars. These vehicles play a crucial role in saving lives and responding to emergencies, but they often face significant setbacks at traffic signals during peak hours, which can drastically reduce their response times. In situations where every second counts, these delays could result in tragic outcomes.

Conventional traffic management systems typically rely on fixed signal timings and, at times, manual intervention to manage the flow of vehicles at intersections. While these systems may work reasonably well under normal traffic conditions, they are increasingly inadequate in dealing with the dynamic and unpredictable nature of urban traffic. The inflexibility of traditional systems makes it difficult to prioritize emergency vehicles effectively, particularly in highly congested urban environments.

However, with the rapid advancements in artificial intelligence (AI) and machine learning (ML), there is now an opportunity to develop more intelligent and adaptive traffic management systems. These next-generation systems can adjust in real time to fluctuating traffic conditions, offering improved efficiency and safety for all road users. One particularly innovative approach is the use of sound-based detection techniques to identify the presence of emergency vehicles. By recognizing distinctive audio signals, such as the sirens from ambulances, fire trucks, and police cars, traffic management systems can be equipped to respond swiftly to emergencies.

Integrating audio classification techniques with existing traffic management infrastructure can provide a significant boost in managing urban mobility. This integration enables the dynamic adjustment of traffic signal timings in favor of emergency vehicles, allowing them to navigate intersections with minimal delay. Such an approach not only ensures faster response times for emergency services but also improves the overall flow of traffic in critical situations.

The system outlined in this paper seeks to leverage machine learning algorithms to classify audio signals from urban environments effectively. The primary goal is to enhance traffic signal control by prioritizing emergency vehicles based on real-time sound recognition. By utilizing advanced techniques such as Mel-Frequency Cepstral Coefficients (MFCC) for feature extraction and Convolutional Neural Networks (CNN) for classification, the system can reliably distinguish between emergency vehicle sirens and various other ambient traffic noises. This capability forms the foundation for an intelligent traffic management solution that addresses the unique

challenges posed by urban congestion, ensuring smoother and faster passage for emergency responders.

II. RELATED WORKS

Intelligent Transportation Systems (ITS) are becoming increasingly important in addressing the growing complexities of urban traffic management. As cities become more populated, the demand for efficient traffic control systems rises, necessitating the incorporation of cutting-edge technologies such as artificial intelligence (AI) and machine learning (ML), fog computing, and edge computing. These technologies have enabled data pre-processing and analysis, leading to significant improvements in traffic flow, congestion management, and overall transportation safety.

One of the key innovations in ITS is the integration of edge computing and deep learning for traffic flow detection. Gu, Hu, and Jia suggests that edge computing handles data nearer to its origin, greatly minimizing latency and facilitating real-time decision-making. This is vital in large urban areas, where timely interventions can prevent bottlenecks. By combining edge computing with deep learning, ITS frameworks can accurately predict traffic patterns and identify potential congestion points before they escalate, enabling traffic authorities to adjust signals and reroute vehicles effectively [3].

In parallel, fog computing enhances traffic control systems by partitioning urban regions into smaller, manageable zones. According to Gu et al., each zone collects locally processed real-time traffic data to dynamically adjust traffic signal timings. This method minimizes delays that may occur when data is sent to distant cloud servers for analysis, making it particularly useful during peak traffic times or sudden congestion due to accidents [3].

The development of smart traffic lights is another significant advancement in ITS. Oliveira, Manera, and Garcez highlight that these systems, integral to smart city initiatives, utilize wireless communication to gather and analyze real-time traffic data. Based on current conditions, smart lights can adjust their timings to optimize vehicle flow through intersections and respond to special events like accidents by rerouting traffic to minimize delays. Such systems aim not only to reduce congestion but also to mitigate environmental impacts by decreasing fuel consumption and emissions through more efficient management [4].

AI-powered traffic information systems have also made notable strides. Gebre et al. explore the integration of Physics-Informed Neural Networks (PINN) and models like GPT-4 to provide real-time traffic monitoring and predictive capabilities. By analyzing extensive datasets, these systems can forecast congestion points and offer solutions, such as rerouting suggestions or signal adjustments. This proactive approach enhances urban mobility, leading to reduced travel times and fewer traffic-related delays [5].

Traffic safety improvements have emerged through the introduction of network-level safety metrics (NSMs). Chen et al. demonstrate that these metrics, derived from deep learning models analyzing roadside video data, can identify potential safety risks, such as unsafe driving behaviors or hazardous intersections. Traffic authorities can leverage these insights to implement targeted safety measures, ultimately reducing accident rates and enhancing overall urban transportation safety [6].

Machine learning algorithms, such as the Artificial Rabbits Optimization (ARO) combined with ensemble learning techniques, have been successfully applied to urban traffic flow monitoring. Ragab et al. indicate that these algorithms analyze complex traffic patterns, enabling accurate predictions of congestion. The integration of ARO ensures effective urban traffic management, allowing authorities to pre-emptively adjust signal timings or suggest alternative routes to reduce congestion [7].

Sustainability is also a critical focus within modern ITS. Chen et al. discuss the Autonomy, Control, and Planning (ACP) framework, which aims to balance traffic flow optimization with energy consumption. By implementing eco-friendly practices such as dynamic lane management and optimized signal control, these systems contribute to a reduction in the overall carbon footprint of urban transportation [8].

In "An Agent-Based Traffic Recommendation System: Revisiting and Revising Urban Traffic Management Strategies," Junchen Jin et al. develop a traffic management framework that leverages agent-based modeling to offer real-time, personalized traffic recommendations. The system represents individual drivers or vehicles as "agents" with unique preferences, behaviors, and objectives, which enables the simulation of complex interactions and behaviors in urban traffic. Through this agent-based approach, the model captures dynamic traffic scenarios, making it possible to provide recommendations that adjust in real-time to minimize congestion and improve travel efficiency across an entire urban network. [9]

In the study "Smart City Traffic Management: Acoustic-Based Vehicle Detection Using Stacking-Based Ensemble Deep Learning Approach," Ahsan Shabbir et al. propose an innovative traffic management system for smart cities that leverages acoustic signals to detect vehicles. This approach is grounded in the stacking-based ensemble deep learning method, which combines multiple model predictions to enhance detection accuracy. By analyzing sound signatures of different vehicles, the model effectively identifies vehicle presence and type, a solution particularly advantageous for real-time monitoring and improved traffic flow management. The paper highlights how this non-visual, cost-effective solution holds promise for urban areas with high traffic density, demonstrating robustness against environmental factors that can impact other sensor types, such as cameras or radar. [10]

Abdulrahman Alruban et al., in their paper "Artificial Hummingbird Optimization Algorithm With Hierarchical Deep Learning for Traffic Management in Intelligent Transportation Systems," introduce a bio-inspired

optimization algorithm for 8 intelligent transportation systems (ITS). The proposed method, the Artificial Hummingbird Optimization Algorithm, is integrated with a hierarchical deep learning model to optimize traffic flow and reduce congestion. By mimicking the foraging behavior of hummingbirds, this algorithm efficiently explores optimal solutions for complex traffic patterns, enabling more adaptive and responsive traffic management. The paper emphasizes the potential of bio-inspired techniques in ITS, highlighting how hierarchical deep learning can be effectively combined with optimization strategies to enhance decision-making in real-time traffic scenarios. [11]

Existing research in sound-based emergency vehicle detection for traffic management faces several limitations. Primarily, studies often rely on limited datasets that lack diverse urban noise conditions, which restricts the generalizability of models to varied environments. Additionally, many approaches focus solely on audio data without integrating visual or radar information, which can reduce accuracy, especially in high-density, noisy settings. Furthermore, the absence of real-time traffic data integration limits the adaptability of these models to dynamic road conditions, hindering their potential to make context-aware adjustments. These challenges suggest that future work should enhance dataset diversity, incorporate multi-modal sensing, and integrate real-time data for more robust and adaptable traffic management solutions.

In summary, the incorporation of cutting-edge technologies such as artificial intelligence (AI) and machine learning (ML), and Convolutional Neural Networks (CNNs) is revolutionizing urban traffic management, particularly in the context of emergency vehicle detection. By leveraging CNNs for sound classification, this approach effectively identifies and distinguishes the unique audio signatures of emergency vehicle sirens, such as those from ambulances, fire trucks, and police cars.

This innovative sound-based detection system enhances traffic flow efficiency by enabling dynamic adjustments to traffic signals, allowing emergency vehicles to navigate intersections with minimal delay. As a result, response times are significantly improved, contributing to better safety outcomes during critical situations. Furthermore, the integration of this technology within Intelligent Transportation Systems (ITS) promotes sustainability by reducing congestion and optimizing traffic patterns.

As urban areas continue to grow, the application of CNNs in emergency vehicle detection will become increasingly vital in creating efficient, safe, and environmentally friendly transportation systems.

III. PROPOSED METHODOLOGY

The development of the emergency vehicle sound classification system was broken down into several key phases: **data collection**, **feature extraction**, **model building**, and **evaluation**. These stages demonstrate the development of a Convolutional Neural Network (CNN)

model designed to classify the sounds of emergency vehicles, such as ambulance, fire truck, and police sirens.

A. Data Collection

The dataset used in this project consists of 600 audio samples in **WAV format**, equally split across three categories: **ambulance sirens**, **fire truck sirens**, and **police sirens** (200 samples for each class). The dataset was uploaded to Google Drive and accessed via **Google Colab** to streamline the training process and leverage cloud resources for model development.

Each audio sample was recorded in various urban environments to introduce background noise such as traffic, human voices, and city ambiance. This diversity was crucial to help the model generalize across real-world conditions, ensuring the robustness of the model when deployed.

B. Feature Extraction

The primary features utilized for sound classification were derived using **Mel-Frequency Cepstral Coefficients (MFCCs)** [11]. MFCCs are a commonly employed feature extraction technique in audio processing, as they accurately represent the short-term power spectrum of sound signals, closely resembling human auditory perception. MFCCs are computed using the following steps:[12]

- **Short-Time Fourier Transform (STFT):**

The STFT is applied to analyze the frequency content of short segments of the audio signal over time, capturing time-frequency characteristics. The STFT of the signal $x[n]$ at time t and frequency f is given by:

$$X(t,f)=\sum_{n=-\infty}^{\infty} x[n]w[n-t]e^{-j2\pi fn} \quad (1)$$

Where:

- $X(t,f)$ is the STFT of the signal at time t and frequency f .
- $x[n]$ is the audio signal.
- $w[n]$ is a window function (like Hamming).

- **Power Spectrum:**

After obtaining the STFT, the power spectrum is computed to analyze the signal's energy distribution across frequencies. This is calculated by taking the squared magnitude of the STFT:

$$P(t,f)=|X(t,f)|^2 \quad (2)$$

This step is essential for understanding the intensity of various frequency components over time, allowing the identification of dominant frequencies in the signal.

• Mel Filter bank:

The power spectrum is then mapped onto the Mel scale, which better reflects human perception of sound. The Mel frequency $M(f)$ corresponding to a physical frequency f is calculated as:

$$M(f) = \frac{2595}{\left(1 + \frac{f}{700}\right)} \quad (3)$$

Where:

- $M(f)$ is the Mel frequency corresponding to the frequency f .

This transformation converts the linear frequency scale to a logarithmic scale, allowing the model to focus on perceptually important frequencies.

• MFCC Calculation:

Mel-Frequency Cepstral Coefficients (MFCCs) are vital in sound recognition tasks because they effectively capture the key audio features, closely mimicking human auditory perception. By mapping the power spectrum of a sound signal onto the Mel scale, MFCCs emphasize perceptually important frequencies while filtering out irrelevant noise. This makes them particularly useful for distinguishing emergency vehicle sounds like ambulance, fire truck, and police sirens in noisy urban environments. The extraction process involves applying Short-Time Fourier Transform (STFT), Mel filter banks, and Discrete Cosine Transform (DCT), resulting in a compact representation of the audio data that highlights its spectral and temporal characteristics. As input features for the Convolutional Neural Network (CNN), MFCCs help the model learn distinctive patterns in the siren sounds, reducing computational complexity and improving classification accuracy in real-time applications.

Finally, the Mel-filtered power spectrum is transformed using a Discrete Cosine Transform (DCT) to produce the MFCCs, which represent the shape of the audio spectrum. The MFCC at index n is calculated as:

$$MFCC(n) = \sum_{k=1}^K \log \log(M(k)) \cdot \cos\left(\frac{\pi n(k-0.5)}{K}\right) \quad (4)$$

Where:

- n is the index of the coefficient.
- K is the number of Mel filters.

These steps enable the extraction of features that capture both the spectral and temporal characteristics of the sound, making MFCCs a valuable input for sound classification tasks.

C. Spectral Subtraction

To enhance the quality of the audio signal and isolate emergency sirens from background noise, spectral subtraction is employed as a noise reduction technique. This approach involves analyzing the frequency spectrum of the recorded audio and subtracting an estimated noise spectrum from it, thereby minimizing the impact of irrelevant ambient sounds. Spectral subtraction is particularly effective in environments with steady or predictable background noise, as it enhances the prominence of the siren signals, making them more distinguishable. By applying this method, the system ensures that critical features of emergency sirens remain clear and detectable, while reducing interference from sounds like traffic noise, honking, and engine idling. This noise-reduced signal is then passed on to the feature extraction module for further processing.

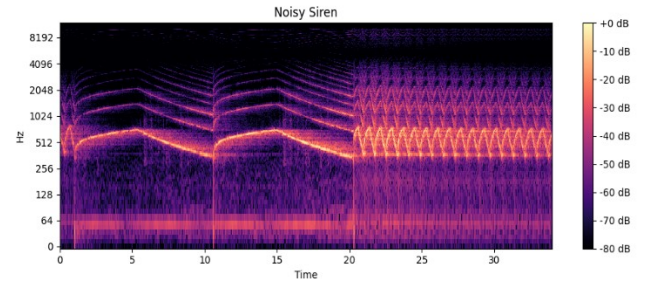


Fig. 1 Noisy Siren

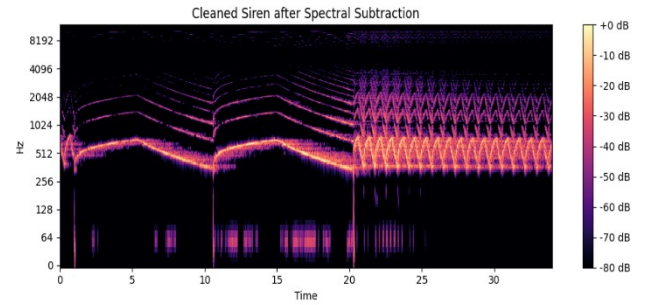


Fig. 2 Cleaned Siren after Spectral Subtraction

D. Convolutional Neural Network (CNN)

A **Convolutional Neural Network (CNN)** was employed to classify the extracted MFCC features into three categories: ambulance, fire truck, and police sirens. CNNs are well-suited for this task as they can learn spatial hierarchies from the data, which, in this case, are the temporal and frequency patterns within the audio signals.

CNN Architecture:

1. Input Layer

The input layer takes in the MFCC features extracted from the audio signals. These features represent the audio in a way that captures important characteristics of the sound, allowing the CNN to analyze the temporal and frequency patterns effectively.

2. Convolutional Layers

Convolutional layers are fundamental to CNNs. They apply convolutional filters (kernels) to the input data to detect local patterns. In the case of audio classification, these patterns could be specific frequencies or temporal changes in the sirens.

3. MaxPooling Layers

MaxPooling layers are used to down-sample the data by taking the maximum value from a set of features within a defined window.

- **Reduced Computational Complexity:** By cutting down on the number of parameters and simplify the calculations, the model speeds up and runs more smoothly.
- **Feature Invariance:** MaxPooling helps in making the model more robust to slight variations in the input, allowing it to recognize the same siren sound even if there are small differences in pitch or volume.

4. Dropout Layers

During training, a certain percentage of neurons in the dropout layer are randomly deactivated, which **Improves Generalization**. Dropout helps the model avoid depending too much on a single feature, which pushes the network to learn a stronger and better-rounded understanding of the data.

5. Flatten Layer

The Flatten layer flattens the 2D feature maps resulting from applying the convolution and pooling layers, turning them into a 1D vector. This layer is needed to get the data ready for what comes next - the fully connected layers.

6. Dense Layer

Dense layers are actually where every neuron is connected to all neurons in the previous layer. Such a structure enables the model to learn complex patterns and reach final predictions or classifications.

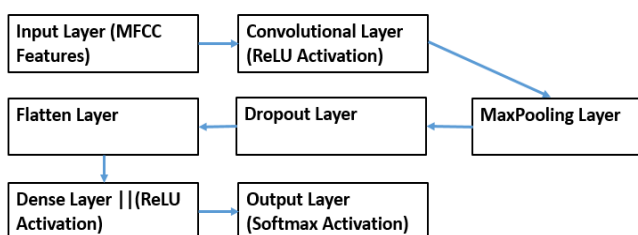


Fig. 3 CNN Architecture Layer

The proposed methodology outlines a systematic approach to developing and optimizing a CNN-based emergency vehicle sound recognition system integrated with intelligent traffic management. The methodology is divided into three primary phases: Sound Recognition Process, Traffic Management Integration, and Model Optimization Techniques.

The sound recognition process begins with audio preprocessing, where noise reduction techniques such as spectral subtraction are applied. This step removes unwanted background noise, including traffic and environmental sounds, enhancing the quality of the recorded emergency siren audio. By ensuring that the input audio data is clean and relevant, the system improves the reliability of subsequent feature extraction and classification stages.

Following preprocessing, the system performs feature extraction using Mel-Frequency Cepstral Coefficients (MFCCs). MFCCs are particularly effective as they capture both spectral and temporal features of sound, closely mirroring human auditory perception. The extracted MFCCs generate a 2D time-frequency representation, which is then fed into a Convolutional Neural Network (CNN) for classification. The CNN processes the MFCC input, learning distinctive patterns and variations across the temporal and frequency domains that are unique to each emergency vehicle siren type, including ambulances, fire trucks, and police vehicles. The CNN classifies the input sound into one of the predefined categories, ensuring robust and accurate recognition. This pipeline ensures that emergency vehicle sounds are detected in real time, enabling timely responses. Once the emergency vehicle sound is recognized, the system integrates the output with urban traffic signal controllers to dynamically adjust traffic signal timings. Upon detection of an emergency vehicle sound (ambulance, fire truck, or police siren), the system immediately communicates with the traffic controller to prioritize the emergency vehicle. The recognition of the siren triggers a priority signal, creating a green corridor by altering standard traffic light timings. This real-time response significantly reduces delays and ensures that the emergency vehicle passes through intersections with minimal waiting time. By dynamically adjusting the traffic flow, the system not only improves emergency response times but also enhances overall urban traffic efficiency. This intelligent approach minimizes congestion and reduces the impact on other road users, contributing to safer and more efficient transportation systems. To achieve high accuracy and precision in classifying emergency vehicle sounds, several optimization strategies are implemented. First, data augmentation and hyperparameter tuning are employed to improve the performance of the CNN model. Data augmentation increases the diversity of the training dataset, enabling the model to generalize better across varying urban noise conditions. Cross-validation is also applied to ensure reliable and robust performance evaluation. To achieve high precision, Transfer Learning is utilized by leveraging a pre-trained model. The pre-trained model initializes the CNN's weights using knowledge gained from large-scale audio datasets, reducing the need for extensive training data. Fine-tuning the pre-trained model on the specific emergency siren dataset further enhances its ability to distinguish between subtle sound differences, improving

classification precision. Additionally, improving the quality and diversity of the training dataset ensures the model performs effectively in diverse urban scenarios. For future enhancements, Ensemble Learning techniques can be explored, though they may require higher computational resources.

Efficiency is crucial for real-time applications like dynamic traffic signal control. The model's computational complexity can be reduced by incorporating techniques such as **MaxPooling layers**, which down-sample the feature maps, decreasing the number of parameters and speeding up processing without sacrificing important features. Furthermore, **Dropout layers** are used during training to prevent overfitting by randomly deactivating a fraction of neurons. This encourages the model to learn more robust features, making it less prone to overfitting and enhancing its reliability in real-world scenarios. Together, these techniques ensure that the model not only achieves high accuracy and precision but also operates efficiently in real-time environments.

IV. RESULTS AND DISCUSSION

The dataset comprised 600 audio samples of emergency vehicle sirens, including ambulance, fire truck, and police sirens. To evaluate the model's generalization ability, the dataset was divided into two parts: 80% training and 20% testing. This corresponds to 480 samples used for training the Convolutional Neural Network (CNN) model and 120 samples reserved for testing its performance on unseen data. The 80-20 data split helps make sure the model doesn't over fit and gives a good idea of how it will perform in real-world situations. By splitting the data this way, we can see how well the model learns from the data it knows and how well it handles new, unseen examples.

The **Librosa** library in Python was used to extract **40 MFCC coefficients** from each audio sample. These coefficients were normalized and averaged to standardize the input for the CNN model.

Steps in Feature Extraction:

1. Load each **WAV** file.
2. Extract 40 MFCC coefficients using the **Librosa** library.
3. Normalize and scale the MFCC values to prepare them for model input.

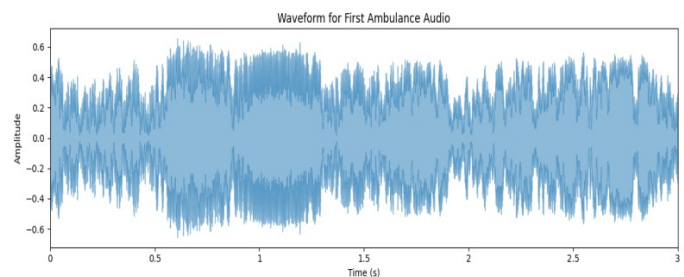


Fig. 4 Waveform for First Ambulance Audio

This transformation allowed the CNN to capture the nuanced differences between the siren types in the temporal and frequency domains. To better understand the frequency-domain differences among the three classes (ambulance, fire truck, and police sirens), spectrograms were generated for each sample. Upon visual inspection, the spectrograms revealed clear patterns unique to each type of siren. For instance:

- Ambulance Sirens: Displayed periodic frequency modulations, alternating between high and low frequencies.
- Fire Truck Sirens: Showed more continuous frequency sweeps with less abrupt changes.
- Police Sirens: Had sharp, repetitive peaks with unique high-frequency components.

These unique frequency characteristics were effectively captured by the Mel-Frequency Cepstral Coefficients (MFCCs) used as input features for the CNN. The distinct differences in the spectrograms made it easier for the model to classify the sounds accurately.

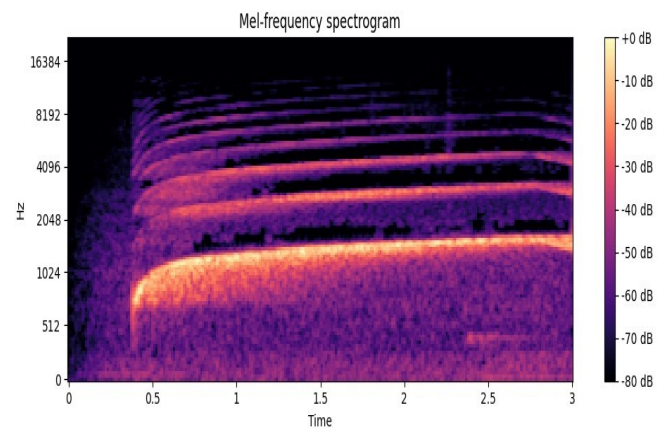


Fig. 5 Mel-frequency Spectrogram

C. Training and Validation Accuracy

During training, both the training accuracy and validation accuracy were tracked to monitor the model's learning progress. The model exhibited a steady increase in both metrics over multiple epochs, indicating that it was learning effectively:

- **Training Accuracy:** Increased consistently as the model became better at fitting the training data.
- **Validation Accuracy:** Showed a gradual rise, the validation accuracy was remarkably close to the curve for training accuracy, showing that the model was indeed not overfitting but simply doing a really good job at generalizing to new, unseen data.

Simultaneously, the loss function—a measure of how far the predicted output deviated from the actual labels—steadily decreased for both the training and validation sets. The drop in loss further proves that the model was successfully learning the underlying patterns of the sirens.

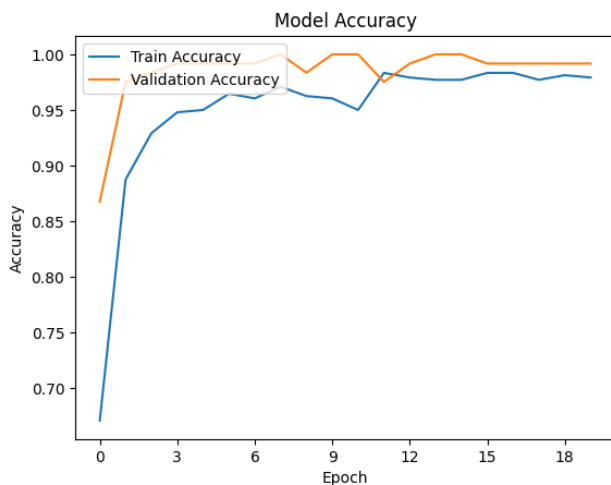


Fig. 6. Model Accuracy

Fig.3. shows the graph represents the model's accuracy over the course of training across 20 epochs. The accuracy is plotted for both the **training** and the **validation set**.

D. Confusion Matrix Analysis

To check how well the model classified the three types of sirens—ambulance, fire truck, and police—a confusion matrix was created. This gives a clear breakdown of how accurately the model predicted each class compared to the actual labels. The confusion matrix highlighted the following:

- The model was most accurate in identifying ambulance sirens, which showed the highest true positive rate.
- Fire truck sirens had some misclassification, mainly confused with police sirens, due to similar frequency modulations in certain scenarios.
- Police sirens exhibited slight misclassifications as fire truck sirens but overall performed well.

This confusion matrix analysis provided insights into the specific areas where the model struggled, guiding future improvements by fine-tuning the model to address these confusions.

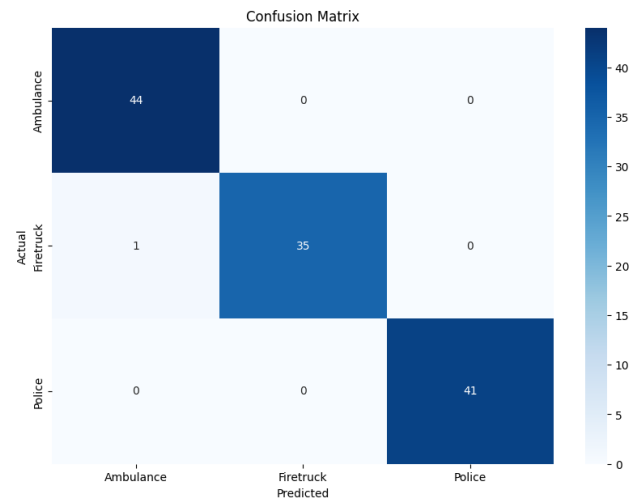


Fig. 7 Confusion Matrix Analysis

Fig.4. shows the confusion matrix indicates that the model performs effectively in recognizing emergency vehicle sirens, with a few misclassifications. This analysis will help guide further improvements and fine-tuning of the model to enhance its classification accuracy, particularly in distinguishing similar sounds.

E. Box Plot Analysis for Loss

Box plots were created to visually assess the distribution of both training and validation loss across all epochs. Box plots offer a statistical summary, including the median, interquartile range (IQR), and potential outliers:

- **Training Loss:** The box plot revealed a relatively low interquartile range (IQR), indicating that the model performed consistently across the training dataset. The minimal spread suggests stable and effective learning.
- **Validation Loss:** Although the validation loss showed slightly higher variability, with a wider IQR, the overall pattern remained relatively stable. The presence of a few outliers in validation loss might suggest some difficult-to-classify samples or temporary fluctuations in model performance during certain epochs.

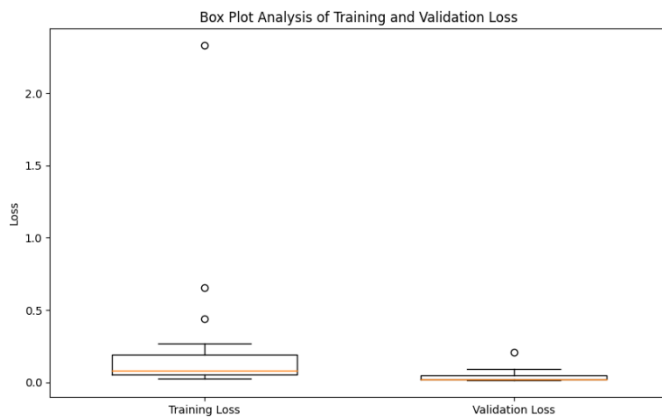


Fig. 8.Box Plot Analysis

V. CONCLUSION

The increasing complexity of urban traffic congestion, particularly regarding the movement of emergency vehicles, underscores the need for advanced traffic management solutions. This research presents a machine-learning-based system designed to detect emergency vehicles by analyzing sound, specifically through Mel-Frequency Cepstral Coefficients (MFCC) for feature extraction and Convolutional Neural Networks (CNN) for audio classification. The system is engineered to accurately identify siren sounds amidst other urban noises and to dynamically adjust traffic signals, thereby giving priority to emergency vehicles. This capability aims to improve traffic flow, minimize delays, and ultimately reduce response times for emergency services. The system has some practical limitations. One constraint lies in the training dataset, which consists of only 600 audio samples. Several enhancements could improve the system's performance and robustness. Expanding the audio dataset to include a broader variety of siren types and diverse urban noise profiles could significantly improve generalizability, enabling the model to better recognize emergency vehicles under varied conditions.

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